

Assignment 8: Equations

DO NOT WRITE ON THIS PAPER

Part A

The following is an expression: $(0.5x-5)$. How do you know it is an expression?

Evaluate it at $x=0$ and $x=10$. Show your steps.

Part B

The following is an equation: $0.5x - 5 = 0$. How do you know it is an equation?

Solve the equation for x . Show your work.

Part C

Consider the equation: $ax - b = cx - d$.

Fill in the values for **a**, **b**, **c**, and **d** by rolling dice. Your first roll is **a**, your second roll is **b**, and so on. Solve the equation for x and explain your steps.

Example: if you roll a 6, 4, 2, and 5, your equation would look like $6x - 4 = 2x - 5$

Question 1

Big Question: What are you allowed to do to an equation?

Here is an equation: $0.2x - \frac{1}{13} = 12x - 3$. Below are a few examples of me "doing something" to an equation. Next to each, write **allowed** or **not allowed** and explain your reasoning.

- a. $123(0.2x - \frac{1}{13}) = 123(12x - 3)$
- b. $(0.2x - \frac{1}{13})^7 = (12x - 3)^6$
- c. $(0.2x - \frac{1}{13}) + 4 = 123(12x - 3) - 4$

Question 2 Work

Solve for x. Include an explanation of each step.

$$\frac{x}{5} + 5.3 = -7.2$$

Question 2 Answer

Write your answer
in your Class Notes

Question 3 Work

Solve for y. Include an explanation of each step.

$$-6 = \frac{5y-17}{-3}$$

Question 3 Answer

Write your answer
in your Class Notes

Additional Space